GOOGLE CLOUD AI BUILDER CUP 2026 · RETAIL & COMMERCE

# Taal

Gemini 3.5 Flash plans in an ADK agent. Deterministic guardrails price every rupee and hold out a control group on every play.

**₹9,191** recovered vs doing nothing on one tea lot PROJECTED

*The agent that sells what the forecast says you'll throw away.*


TEAM

**Nandish** · lead

**Rinu** · co-builder


PROBLEM

1. Forecasts and offers live in different tools: one dark store writes stock off while the next runs out.
2. Customers get blanket markdowns, not offers matched to what they buy.
3. Nothing plans to the online sell-by cut-off: 30% of shelf life or 45 days at delivery.[1]

[1] FSSAI advisory to e-commerce FBOs, 3 Dec 2024, file RCD-13/1/2024-Regulatory-FSSAI(E-13150): minimum 30% shelf life or 45 days at delivery. Either-limit reading is Taal's configurable default.

THE PROBLEM · HOW BIG, WHICH RULE, WHO DECIDES

# The forecast sees the write-off coming, but nothing acts on it.

HOW BIG · AS REPORTED

**1.8%**

**of net order value (NOV) is lost to inventory at Blinkit.**

As reported by Inc42, 22 Jul 2026. Per-store line below shows its inputs, from the same article.

**₹17,132 cr ÷ 2,443 stores × 4 × 1.8% ≈ ₹0.5 cr per store a year**

All inventory loss, one quarter annualised. Context for scale, not a Taal result.

THE RULE · AN ADVISORY, NOT LAW

**30% or 45 days**

**of shelf life must remain at delivery of an online order.**

FSSAI advisory to e-commerce food sellers, 3 Dec 2024. Stock can be in date on the shelf yet past its online sell-by, so the deadline can come weeks before expiry.

Either-limit reading is Taal's configurable default.

WHO IT IS FOR

## A category or supply-chain head

They own write-off and are the buyer. Day to day, two people use Taal:

**A demand planner** checks the gap and the guardrails, then approves the play on the Play Desk.

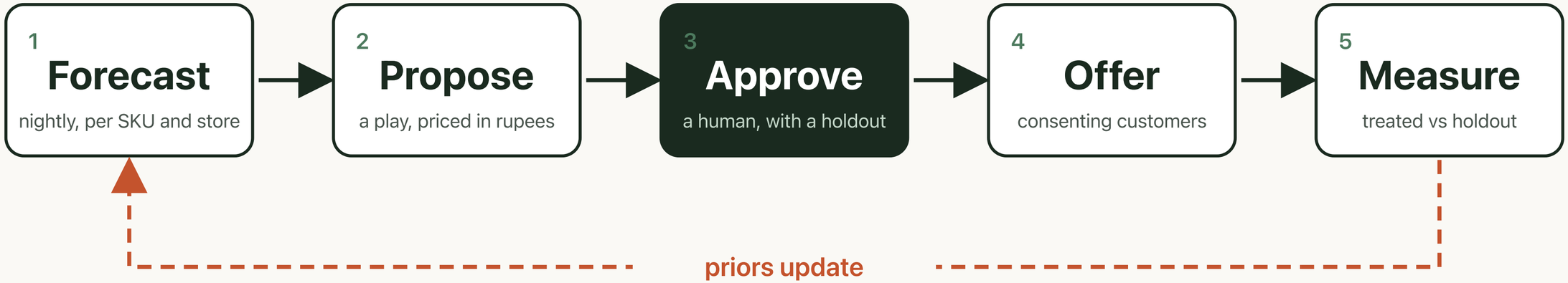
**A dark-store manager** confirms what is on the pallet on the phone view and receives any transfer.

NEXT The loop and the hero result are on slide 3; projected impact is on slide 15; 3 of 7 practitioners we asked write off weekly or daily (slide 13).

THE LOOP AND THE HERO PLAY

# ₹9,191 recovered vs doing nothing on one tea lot, ₹6,186 ahead of a blanket markdown

PROJECTED · SEEDED TENANT




TEA · GEMINI RUN · PROJECTED

**₹9,191**

recovered vs doing nothing (₹9,397 waste avoided – ₹206 transfer); ₹6,186 ahead of a blanket 20% markdown

HOLDOUT · SYNTHETIC ORDERS

**+0.3 pp**

pp = percentage points. 95% confidence interval (CI) −8.7 to +1.6. 1 of 355 treated responded, 0 of 39 held out. SYNTHETIC, inconclusive. Chips, 323 / 29 (19 Sep): also inconclusive

WHAT THIS PROVES

**The loop closes**

Measure runs end to end on synthetic orders; effect size needs a real pilot

THEME Taal forecasts stock per SKU and node nightly, with festival effects, and drafts human-approved offers.

DIFFERENTIATION

# A holdout on every play. The play written back into the forecast.

| | ONLINE SELL-BY CUT-OFF | HOLDOUT PER PROMO | PROMO → FORECAST WRITE-BACK | INDIA / Q-COMMERCE |
|---|---|---|---|---|
| **Taal** | Yes | Yes | Yes | Built for it; seeded tenant |
| **Blue Yonder** | Unclear | Unclear | Unclear | Partial |
| **RELEX** | Partial | Unclear | Partial | Partial |
| **Wasteless** | No | Unclear | Unclear | Unclear |
| **Afresh** | No | Unclear | Unclear | No |
| **Flashfood** | No | No | No | No |

**Unclear** = no public evidence found in vendor pages read on 2026-10-09; it does not mean the feature is absent. Partial and No are our reading of those pages. Afresh and Flashfood are different product categories.

**New: the model proposes, code owns every rupee, and every play keeps a holdout.**

LIST OF FEATURES

# Six capabilities, gated by rules a model never touches.

### Vision intake

Photograph a pallet; dates and counts come back with confidence. Low confidence goes to a person.

### Nightly Sense

Per-SKU, per-node forecasts. Six gap types, each with rupees and a deadline.

### Planner Agent

ADK LoopAgent, max 3 iterations, six deterministic tools; a Cost Governor picks the gaps.

### Approve + assignment

A human approves. Holdout by hash; offers go to treated customers only.

### Customer Agent

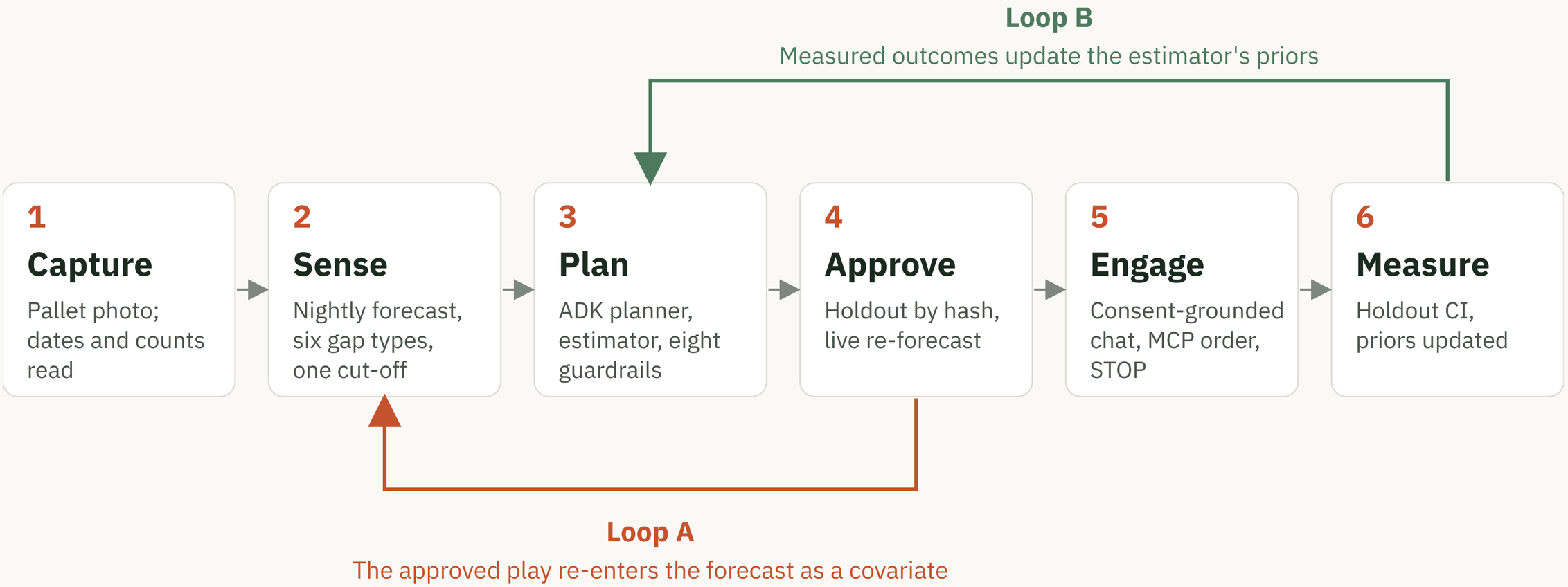
Grounded in stock, offers and consent. Orders through an in-process MCP tool.

### Measure

Treated vs holdout, 95% CI. Priors update from the counts.

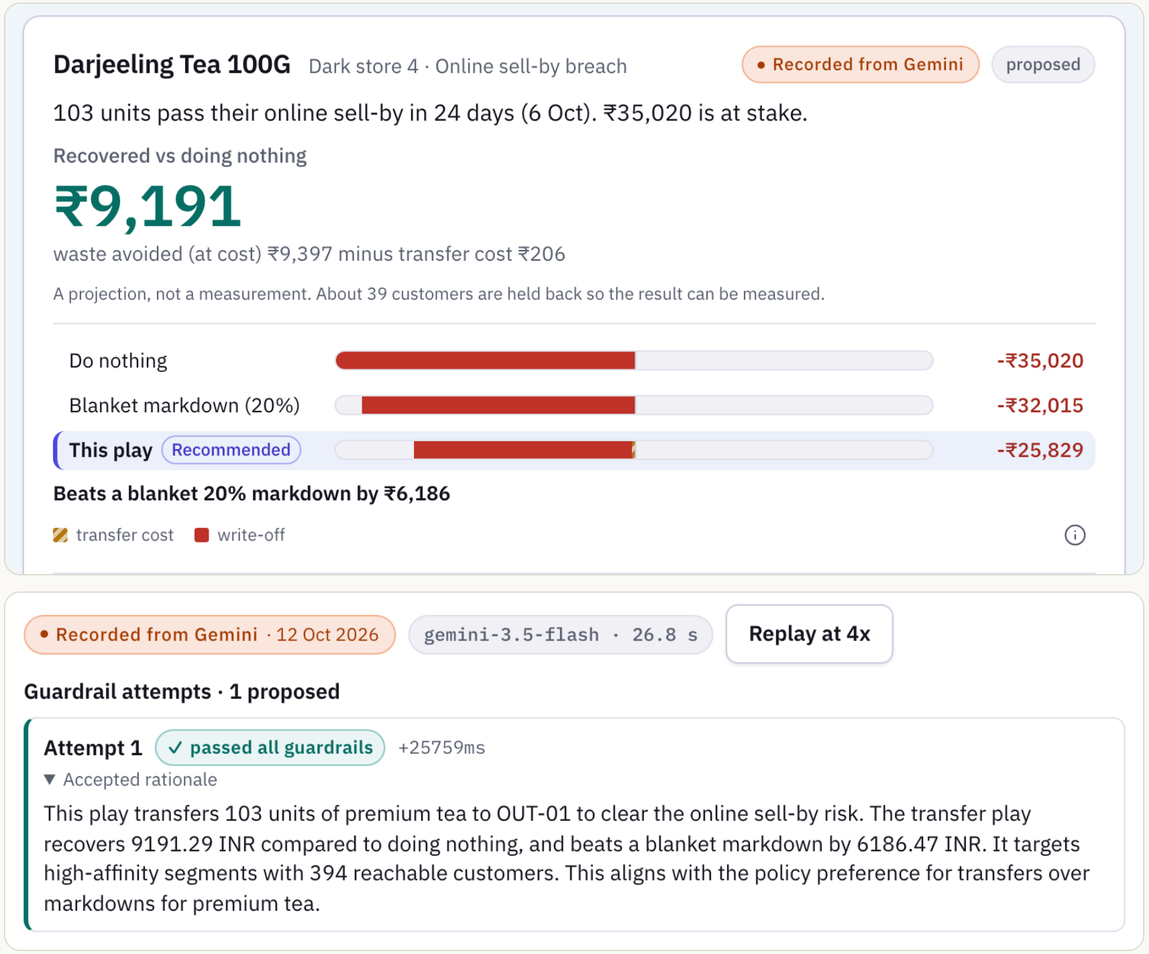
PROCESS FLOW

# Two closed loops, not an open-ended pipeline.



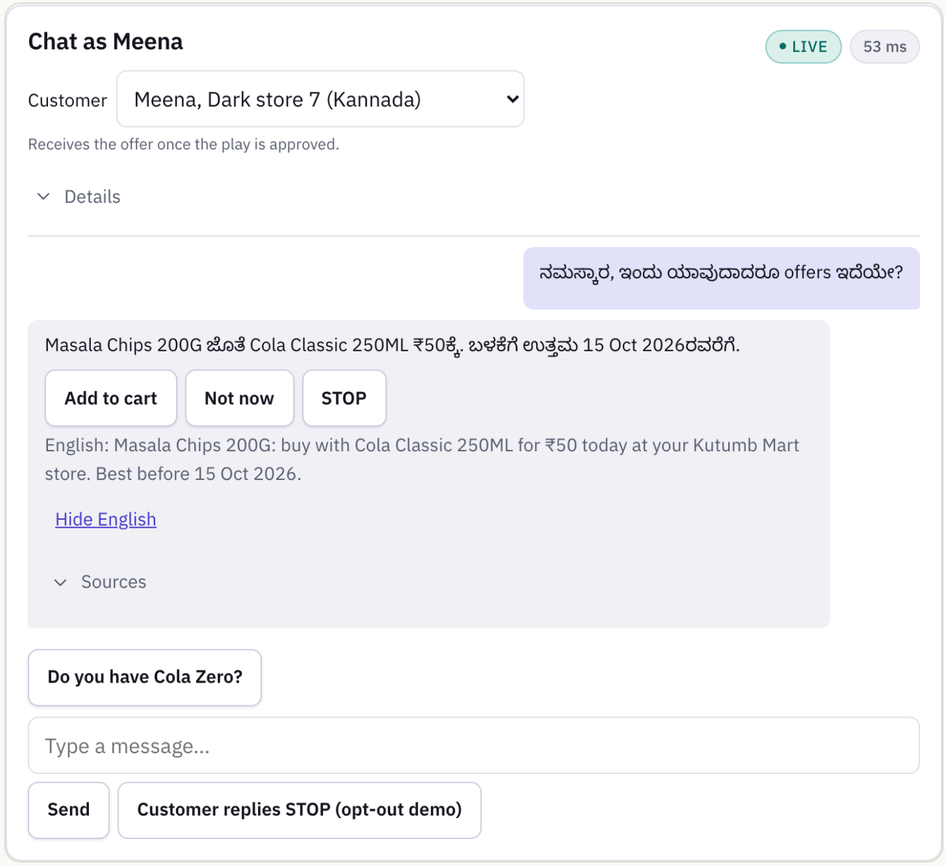
SCREENS

# Gemini drafts the play; code prices every rupee; customers get Kannada plus English.




## Play Desk: the tea play, recorded from Gemini, 12 Oct

Gemini drafted the play and its rationale; code priced every rupee and the guardrails passed it on the first attempt. Projected, seeded tenant.

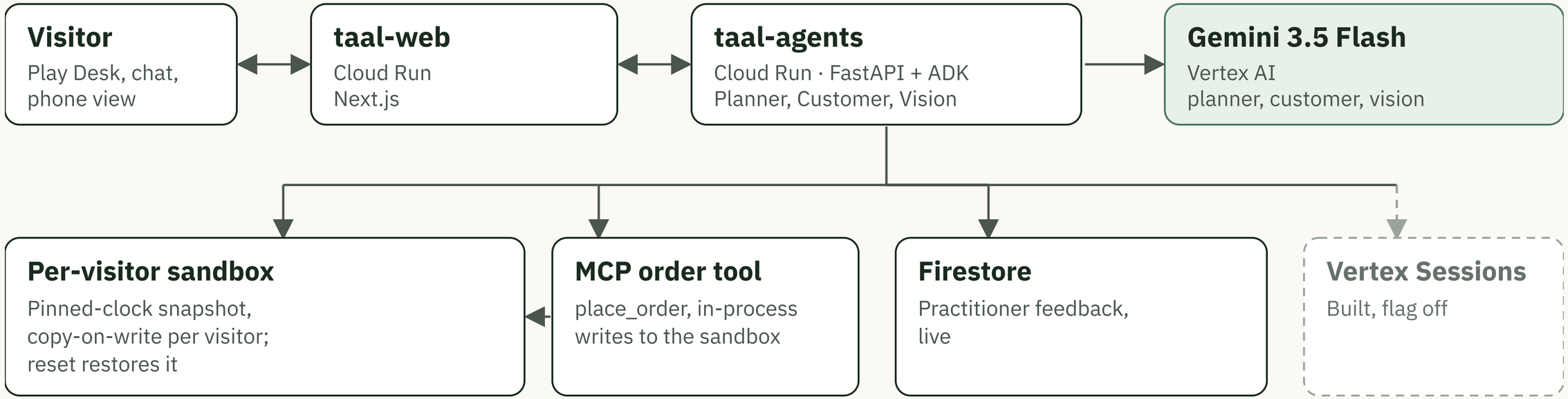



## Customer chat, live

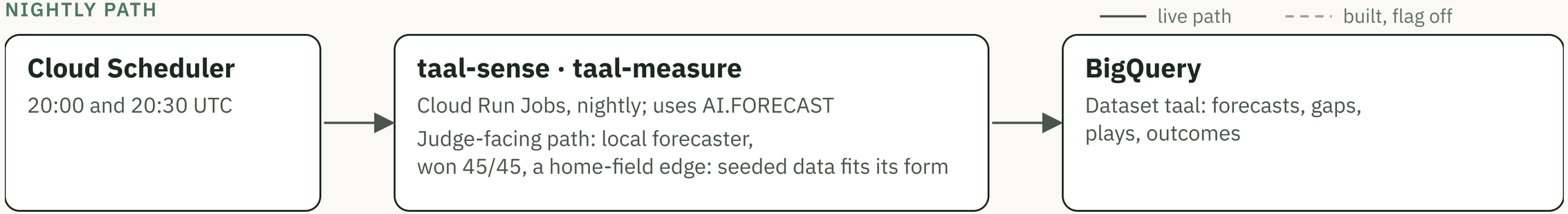
Kannada reply with an English gloss; the order goes through the MCP order tool.

ARCHITECTURE

# A sandbox serves judges; nightly jobs run on BigQuery.




NIGHTLY PATH



TECHNOLOGIES USED

# Each Google service, and the decision it makes.

| SERVICE | THE DECISION IT MAKES |
|---|---|
| **Google ADK planner (Gemini 3.5 Flash)** | Proposes the play: mechanic, audience, window, in a LoopAgent (max 3) with 6 tools |
| **Gemini on Vertex AI** | Drafts the offer, chats with the customer, reads the pallet photo |
| **BigQuery AI.GENERATE_TABLE** | Approve asks it for the offer copy under a 3 s budget; if it misses, templated copy is used. The API reports which (copy.source). |
| **Local forecaster (BigQuery AI.FORECAST nightly)** | Re-forecasts the write-off on Approve; the nightly batch feeds Sense |
| **Firestore** | Stores practitioner feedback, live in the deployed service |
| **Cloud Run, Jobs and Scheduler** | Serves the app and API; runs Sense and Measure nightly; scales to zero |
| **MCP order tool** | Places the customer's order when they accept the offer |
| **Deterministic guardrails and estimator** | Check every play against 8 rules and own every rupee; Gemini never sets a number |

DEPLOY GitHub Actions runs verify, then deploys to Cloud Run on every merge to main.

ESTIMATED IMPLEMENTATION COST

# Priced from measured usage; sold on rupees rescued.

PER 30 DAYS, MODELLED FROM MEASURED USAGE

## ₹486 / $5.86

**Measured 23 Sep on gemini-2.5-flash, before the 10 Oct switch to gemini-3.5-flash. Not re-measured.**

Measured with no plays in the window: idle-plus-nightly cost.

Real Gemini tokens, 752 BigQuery jobs and Cloud Run time, priced at public list rates. Not a bill.

*Method: modeled_from_measured_usage.*

### Forward-looking profiles ESTIMATES

| Profile | Scale | Monthly |
|---|---|---|
| **Micro** | 1 shop, ≤500 SKUs | ₹200–500 |
| **Mid** | 300 SKUs, 10 nodes | ₹2,500–5,000 |
| **Large** | 20,000 SKUs, 500 nodes | ₹1–2 lakh |

BUYER AND PRICING

**Buyer:** the category or supply-chain head, who owns write-off.
**Basis:** a fee per node per month, or a percentage of rupees rescued vs the holdout.

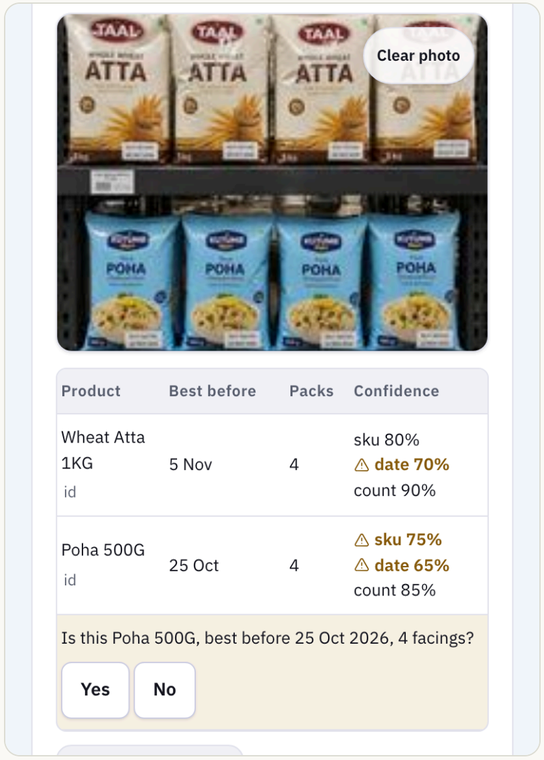
PROJECTED MARGIN LESS RUN COST ILLUSTRATION

**≈ ₹21,710 – ₹156 to ₹313 per node**

Per 28 days at the 5% prior: ₹347,356 margin over 16 nodes, less Mid-profile cost (₹2,500–5,000 a month, estimate). The ₹486 had no plays in its window. Synthetic.

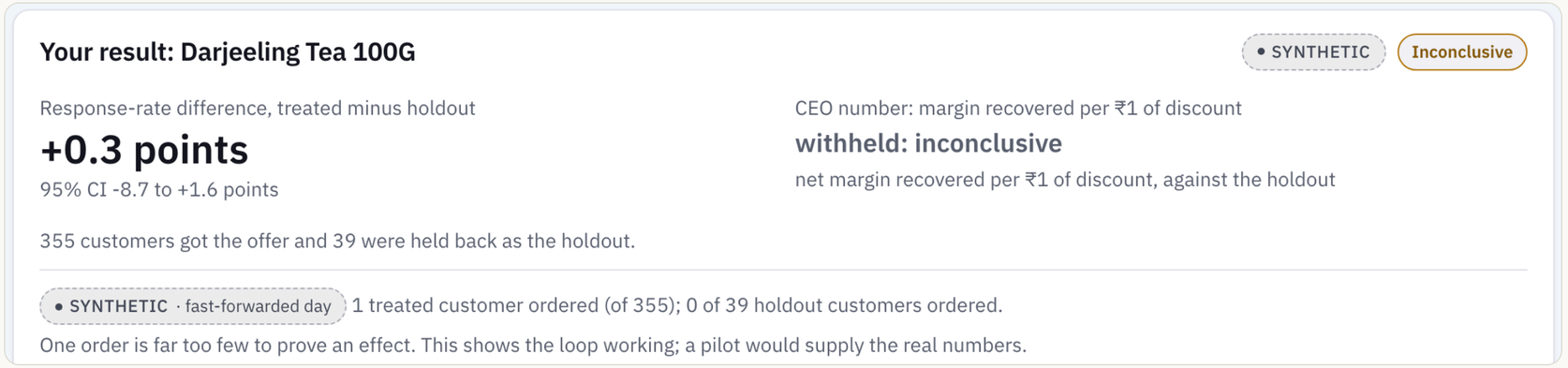
SNAPSHOTS OF THE PROTOTYPE

# Doubtful reads go to a person; outcomes say SYNTHETIC.




## Phone view

A read under 70% confidence is asked back before it reaches inventory.




## Outcomes on a fast-forwarded day, synthetic orders: Tea

355 offered, 39 held out; 1 treated customer ordered, 0 held-out. Response rate +0.3 percentage points, 95% confidence interval −8.7 to +1.6. The range crosses zero, so the page says inconclusive and withholds the rupee figure.

PERFORMANCE REPORT

# Measured and dated; each card names its source file.

### PLANNER, LIVE — 10 Oct

**5/5**

**gemini-3.5-flash · chips**

proposed, 0 fallbacks; 21.8–27.6 s each. Tea, 12 Oct: 8/8, 19.0–31.4 s

`planner_real_traces_2026-10-10/ and _tea_2026-10-12c/ summary.json`

### CUSTOMER AGENT — 28 Sep

**4.56 s p50**

**gemini-2.5-flash**

p95 6.48 s, 0.48 s over the 6 s budget; 50/50 returned 200

`eval/raw/customer_latency_2026-09-28/analysis.json`

### APPROVE · CURL, 6 CALLS — 12 Oct

**about 3–7 s live**

**after concurrent copy + re-forecast**

after #90; first call after idle is slower. Sep: 8.3, 10.6 s

`eval/raw/approve_timing_2026-10-12b.json`

### AGENT SIMULATION — 21 Sep

**190/190**

**gemini-2.5-flash**

live Vertex; stock, holdout arm, stacking, STOP, hallucination

`eval/raw/agent_simulation_2026-09-21.json`

### VISION INTAKE — 21 Sep

**30/30**

**gemini-2.5-flash**

correct dates; synthetic photos only

`eval/raw/vision_synthetic_2026-09-21.json`

### BACKTEST, SYNTHETIC — 20 Sep

**0.113 WAPE**

**statistical forecaster, no LLM**

MAPE 0.133 over 63 tier-origin rows

`eval/raw/backtest_rows_2026-09-20.jsonl · eval/raw/external_backtest_2026-10-09.json`

### PUBLIC NON-GROCERY DATASET BACKTEST — UCI Online Retail II

**0.83 WAPE**

**statistical forecaster, no LLM**

Method and caveats in eval/external_backtest.md

`eval/raw/external_backtest_2026-10-09.json`

### LIVE PLAN, DEPLOYED · GEMINI-3.5-FLASH — 11 Oct

**22/24 = 92% passed**

model-planned, 0 fallbacks; planner p50 26.5 s, p95 31.8 s. 2 polls hit a cold instance, counted as not passed

**47.7 / 47.7 / 30.8 s**

before the fix, 9 Oct: 2 of 3 hit a 45 s deadline and fell back to rules

`eval/raw/plan_live_2026-10-12.json · eval/raw/live_replan_timing_2026-10-09.json`

### ADK EVAL, VERTEX — 12 Oct

**43/46**

**response match (93%)** gemini-3.5-flash · 12 Oct · prompt v6. 38/46 on gemini-2.5-flash, 9 Oct

**1/46**

**exact tool trajectory** (6/46 on 9 Oct): recordings contain retries, so exact order differs. **First proposed play passed the gate 41/46** (5/46 on 2.5-flash, 9 Oct)

**15/15**

**rationale judge**, same-family judge. Prompt v7 (shipped Tea play) not measured

`eval/raw/live_eval_2026-10-12/`

**Tests:** 470+ pytest tests and 490+ Playwright tests run in **make verify**, the CI gate (.github/workflows/verify.yml) on every pull request and every push to main; a second CI job runs 24 live-stack tests. **Models:** cards tagged gemini-2.5-flash predate the 10 Oct switch to gemini-3.5-flash (2.5 retires 20 Oct) and have not been re-run.

VALIDATION · PRACTITIONER RESPONSES, SEPTEMBER 2026

# What seven practitioners told us.

**3 of 7**
write off stock weekly or daily

**3 of 6**
online sellers say the sell-by rule limits what they sell

**0 of 7**
measure a promotion against a control group

**4 of 7**
want a one-week pilot

*"Blanket markdowns erode margin on products that may still sell normally."*

Category / merchandising, quick-commerce

*"Item-store recommendations with incremental-margin estimates."*

Category / merchandising, quick-commerce

7 usable responses (10 collected, 3 excluded as flagged in config/feedback_exclusions.json). Anonymous by consent; contacts never published. Counts, not percentages.

USER GUIDE

# Three people, three screens, one approved play.

## Planner

/desk

1. Open the Play Desk and pick a gap.
2. Check the guardrails and counterfactuals.
3. Click Approve. The forecast moves.

## Node manager

/phone

1. Photograph a pallet, or tap a sample.
2. Confirm the rows read below 0.7 confidence.
3. Approve the play from the gap card.

## Customer

/chat

1. Open the chat. It starts with "Any offers today?".
2. Ask about a product. You see a stock band, not a count.
3. Send STOP to withdraw consent.

**Badges.** LIVE: a real call this visit. REPLAY: stored output replayed. SYNTHETIC: fabricated demo data. "Recorded from Gemini": a dated real model run.

IMPACT · WHAT IS MEASURED, WHAT IS PROJECTED

# Projected margin hinges on one assumption.

TWO LOTS · PROJECTED

### Tea (recorded Gemini run)

**₹9,191 recovered**

**Vs doing nothing; projected, seeded tenant. ₹6,186** ahead of a blanket 20% markdown. 103 units, ₹35,020 at stake.

### Chips (10 Oct, API snapshot)

**-₹279**

Projected. 368 units, ₹9,200 at stake. Net −₹8,468 vs −₹8,189.

RESPONSE PRIOR SENSITIVITY · PROJECTED

| PRIOR | MARGIN | WASTE AVOIDED |
|---|---|---|
| 2% | ₹1.41 lakh | ₹3.26 lakh |
| **5% default** | **₹3.47 lakh** | **₹7.06 lakh** |
| 10% | ₹6.19 lakh | ₹11.18 lakh |

Seeded tenant, 28-day horizon, 550 grocery gaps; plays planned at each prior: 421 (2%), 424 (5%), 423 (10%).

MEASURED VS PROJECTED

**Measured on real customers:** none yet.

**Synthetic holdout, Tea:** 355 treated, 39 held back; 1 responder vs 0, inconclusive.

**Synthetic holdout, Chips (19 Sep):** 323 treated, 29 held back; +1.5 points, 95% CI −12.4 to +5.6, crosses zero.

**Projected at 5%:** ₹347,356 margin, ₹706,183 waste avoided.

**Practitioners:** 7 usable responses; 3 of 7 write off stock weekly or daily.
**Scale of the problem:** sourced loss figure and arithmetic on slide 2 (as reported, all inventory loss, not a Taal result).

LONGEVITY · RUNNING IT FOR REAL

# What it takes to run beyond the demo.

## Production path

**Today:** services run as the owner-role deploy identity; CI uses a stored key. Secrets sit in a managed store.

**Next:** least-privilege service accounts and workload identity.

Model IDs pinned with retirement dates: gemini-3.5-flash live, Flash-Lite fallback; the previous model retires 20 Oct 2026.

## Scalability

**Measured:** Sense 3.8 s (local, 300 SKUs); chat p50 4.6 s, p95 6.5 s (50 calls, 28 Sep, gemini-2.5-flash); live plan p50 26.5 s, p95 31.8 s (22/24, 11 Oct); Approve about 3 to 7 s (measured 12 Oct after #90; first call after idle takes longer).

**Estimate:** 20,000 SKUs, 500 nodes. ESTIMATE

Breaks first: max-instances 1 with re-plan state in memory, then the planner queue.

## Feasibility

Buyer: category or supply-chain head. Fee per node or rupee share.

Pilot: 50 to 60 testers, one week, not yet run.

Run cost: ₹486 a month, usage measured 23 Sep on gemini-2.5-flash, not a bill.

Estimates: ₹200 to 500 (Micro), ₹2,500 to 5,000 (Mid), ₹1 to 2 lakh (Large). ESTIMATE

ADDITIONAL DETAILS · FUTURE DEVELOPMENT

# Next: a real pilot, real photos, more tenants.

1

## A pre-registered pilot

One week, 50–60 consented testers, 40–50% holdout. Metric pre-registered in docs/pilot.md.

DESIGNED

2

## Vision on real photos

Harness built. Real camera photos are the missing input; 30/30 is on synthetic ones.

HARNESS BUILT

3

## Multi-tenant BigQuery

Tenant id on every table; the BigQuery path is built, flag off until a second tenant.

BUILT, FLAG OFF

LINKS

# Taal: the agent that sells what the forecast says you'll throw away

GITHUB REPOSITORY

https://github.com/Nandish3010/probable-fortnight

LIVE APP

https://taal-web-2obkp776ca-el.a.run.app

₹9,191 recovered vs doing nothing on one tea lot · PROJECTED, seeded tenant
22 of 24 live plans on gemini-3.5-flash, p50 26.5 s (11 Oct)
Holdout so far: synthetic and inconclusive; a real pilot is next

Open the live app, click Run the 60-second beat, then Approve. Nothing you do there is saved.

**Team Taal** Google Cloud AI Builder Cup 2026 · Retail & Commerce